BoWatt Technical Test

Objective

Create a small, Instagram-like web application to demonstrate your technical skills in full-stack web development. The site should let users upload images anonymously and view newly uploaded images live in a feed.

Requirements

Frontend (React / TypeScript)

Create a simple, clean, and responsive React/TypeScript-based frontend. The frontend should let users:

- Upload new images with a title and tags.
- See uploaded images with their titles and tags. This feed should be updated live via a websocket.
- Filter the feed by tags.

Backend (Go/Another compiled statically typed language)

Develop an API to handle requests from the frontend. You may use any libraries you see fit. Errors should be handled/reported appropriately. The API should:

- Serve an initial set of images through a REST endpoint.
- Accept uploads over a /uploads endpoint (storing files locally is fine).
- Provide and manage websocket connections for notifying users of newly uploaded images.

Submission Requirements

AI: Use of AI/LLM code generation when solving this technical test is forbidden. We need to see your own abilities to solve issues and find solutions. All code submitted is your responsibility.

Codebase: Provide a link to any repos created for your work.

Documentation:

1. Instructions to set up and run the project.
2. Provide a brief explanation of the architecture and design decisions. Include any plans you had that you may not have had time for.
3. Simple API documentation.

Happy coding!

Bonus Points

If you have time and feel up to an extra challenge, pick and choose any of the following items to include in your submission:

- Use Docker to containerize the application
- Implement a basic caching mechanism to speed up repeated queries
- Add unit tests for key components
- Queuing for file uploads
- Infinite-scroll feed
- Normalise uploaded images
- Polished UI